

Ailin Huang 黄艾琳

Highlights: Wuhan University | ACM-ICPC🥈 | 300+ Cites | Team Leadership | 5 Years of Experience

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Education

Wuhan University | B.Eng. in Computer Science and Technology | 2019 – 2023

Peking University | National School of Development | Minor in Economics | 2024 – 2026 (Part-time)

Experience

2025.05 - Present ✨ StepFun AI | Algorithm Researcher, Multimodal Intelligence Group

Led a team of 5. Responsible for establishing the full-lifecycle evaluation system for the company's LLMs (Pre/Post-training). Spearheaded the development and maintenance of "Step Evaluation," the company's primary full-stack evaluation platform. Developed and integrated over 200 core benchmarks and provided technical support. Responsible for the evaluation modules within the internal training framework "Steptron" and regularly produced competitive analysis reports.

2025.01 ~ 2025.05 ✨ StepFun AI | Algorithm Researcher, AIGC Group

Iterated CRef capabilities for image generation and developed the "ViStoryBench" for story visualization. Published a **CVPR 2026 paper** (Co-first author & Project Lead).

2023.11~ 2025.1 ✨ StepFun AI | Algorithm Researcher, Agent Group

Explored multimodal models, including LLM data processing, SFT/RL training, and speech model training. Open-sourced the 130B "Step-Audio" model and its technical report. Led the iteration of Rap and singing features.

2023.07~ 2023.11 Megvii Research | Assistant Researcher, AIC Group

Focused on Supervised Fine-Tuning (SFT) for LLMs after conversion to a full-time role. Published a paper in **WACV 2024** (Second author) on image processing.

2021.01 ~ 2023.06 Megvii Research | Algorithm Research Intern, AIC Group (Mentor: Shuchang Zhou)

Full-time on-site internship in Beijing. Research areas included video frame interpolation, neural rendering, CLIP, 3D human body, and Talking Head. Published papers in **MM '22 (First author), IJCAI '23 (Co-first author), and CVPR '23 Highlight (Third author)**.

Research Experience

2025 NextStep-1: Toward Autoregressive Image Generation with Continuous Tokens at Scale — Paper | GitHub

NextStep Team

StepFun's unified multimodal autoregressive understanding and generation work. Utilized a 14B language backbone and a 157M Flow Matching head for end-to-end training. Achieved SOTA in open-source autoregressive image generation and editing. Responsible for training data processing.

2025 ViStoryBench: Comprehensive Benchmark Suite for Story Visualization — Paper | GitHub

Cailin Zhuang†, **Ailin Huang**†, Wei Cheng, Jingwei Wu, Yaoqi Hu, Jiaqi Liao, Hongyuan Wang, Xinyao Liao, Weiwei Cai, Hengyuan Xu, Xuanyang Zhang, Xianfang Zeng, Zhewei Huang, Gang Yu, Chi Zhang — CVPR2026

Collected a diverse script-image dataset for Story Visualization. Developed benchmarks addressing domain-specific issues such as cross/self multi-person ID consistency, cinematic style consistency, and prompt following. Evaluated 27 open/closed-source models.

Role: Co-first author/Project Lead. Responsible for early-stage research, metric design, dataset collection/curation, paper writing, and organizing the benchmark coding development.

2025 Step-Audio: Unified Understanding and Generation in Intelligent Speech Interaction — Paper | GitHub

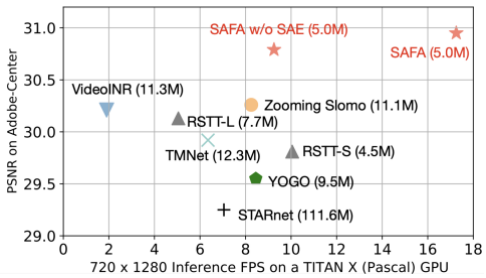
StepAudio Team

Open-sourced a 130B speech-text multimodal large model. Supports fine-grained control with dynamic adjustments for dialects and emotions. Enables conversation, singing, and RAP (Integrated into the "跃问" App). Responsible for implementing RAP and singing functionalities.

2022 Scale-Adaptive Feature Aggregation for Efficient Space-Time Video Super-Resolution — Paper | GitHub

Zhewei Huang, **Ailin Huang**, Xiaotao Hu, Shuchang Zhou — WACV2024

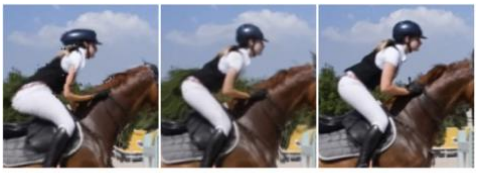
Proposed a scale-adaptive feature aggregation method using optical flow for joint video frame interpolation and super-resolution. SAFA adaptively adjusts feature sizes to suit different scenes. Achieved SOTA on several public benchmarks. Responsible for experiments and paper writing.



2022 A Dynamic Multi-scale Voxel Flow Network for Video Prediction [Paper](#) | [GitHub](#)

Xiaotao Hu, Zhewei Huang, **Ailin Huang**, Shuchang Zhou CVPR2023 **Highlight**

Video prediction using optical flow. We proposed a multi-scale Voxel-Flow block, enabling the model to handle motions across different scales and improving its generalization capability. Experiments demonstrate that during inference, DMVFN adaptively selects the appropriate sub-network based on the input, accelerating inference time while maintaining model performance.
Responsible for a portion of the experimental work and paper writing.



2022 Perceptual Conversational Head Generation with Regularized Driver and Enhanced Renderer [Paper](#) | [GitHub](#)

Ailin Huang†, Zhewei Huang†, Shuchang Zhou ACM-MM2022 Main Track

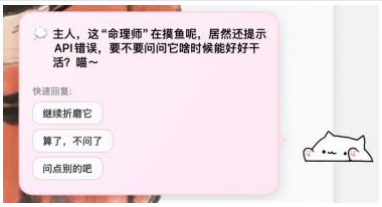
Champion of the ACM Multimedia 2022 Conversational Head Generation Challenge. The competition solution involved training a universal audio-driven motion generator and simulating a high-quality renderer to transform motion representations into video. We conducted fine-tuning on the baseline and utilized a foreground-background fusion module for post-processing the rendered results, while exploring various ensemble learning methods. Responsible for the vast majority of experimental work and paper writing.



Project Experience

2025.10 BongoCat - Step Researcher Edition (Hackathon Winner)

A Swift-based AI desktop pet providing a non-intrusive ambient interaction medium. Leverages StepFun's multimodal capabilities to offer Code Review, Debugging tips, research data queries, and companionship for developers. Uses screen understanding for real-time interaction.



2024.10 Music Generation / Vocal Humming / RAP Tasks

Implemented music MIDI generation and explored low-dimensional representations. Resolved issues in text-to-rap/vocal such as infinite loops, missing words, and style loss. Optimized performance to industry-leading levels.

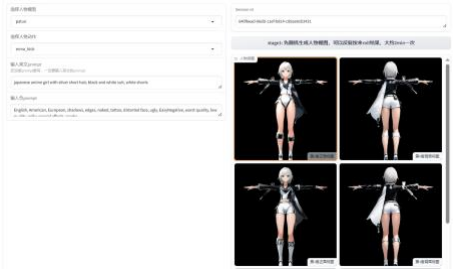
2024.06 Talking Head Generation

Explored various tokens (e.g., 4m21, AU, blendshape, landmark, SVG, live portrait embedding) as facial representations for dimensionality reduction, leveraging large models for expression recognition.
The same data was used for training prediction-based tasks.
Use SVG representation to reduce the dimensionality of facial features.
Trained LLMs to learn the mapping from audio VQ tokens to live portrait representations, achieving voice-to-facial-motion driving.

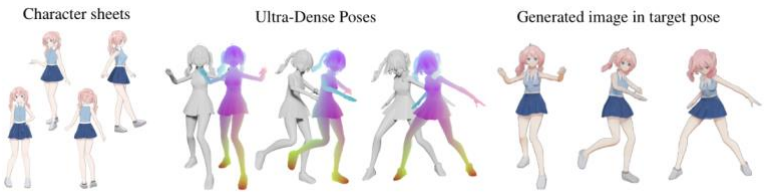


2024.04 Human/Face 3D Art Asset Generation

Generated text from prompts, converted to TTS, then to blendshapes to drive facial expressions. Created high-quality textures from prompts and depth maps to achieve neural-based seamless 3D model re-coloring.



2022.06 Showcased a demo for converting hand-drawn images into animation. The demo takes multiple hand-drawn images of a single character and a motion sequence as input to generate an animated video of the character. Responsible for the production of the demo. GitHub Repo



2021.12 Reimplemented the RIFE (Real-Time Intermediate Flow Estimation) video frame interpolation algorithm using MegEngine and optimized text-driven facial editing algorithms such as StyleCLIP and HyperStyle. Developed a text-driven facial editing pipeline, where the RIFE algorithm was used to blend faces before and after editing to enhance facial similarity.



Honors & Awards

2025	Champion, 2nd StepFun Hackathon (2025)
2022	ACM MM2022 Conversational Head Generation Challenge | Gold Medal (Track 2), Silver Medal (Track 1)
2020 – 2021	**Silver Medalist: ICPC Nanjing (2020), ICPC Yinchuan (2021), CCPC Weihai (2020)**
2020	1st Place (Preliminary), Megvii AI Open Source Competition (2020)
2018	First Prize, National Olympiad in Informatics in Provinces (NOIP, 2018)

Others

Part-time job	Problem Setter for ICPC Kunming Regional 2021; 100+ hours of experience in competitive programming coaching.
Interests	Snowboarding, swimming, board games, raising a Ragdoll cat, and frequent bouldering.
Languages	English (CET-6), Japanese (N3 level equivalent, currently preparing for JLPT).